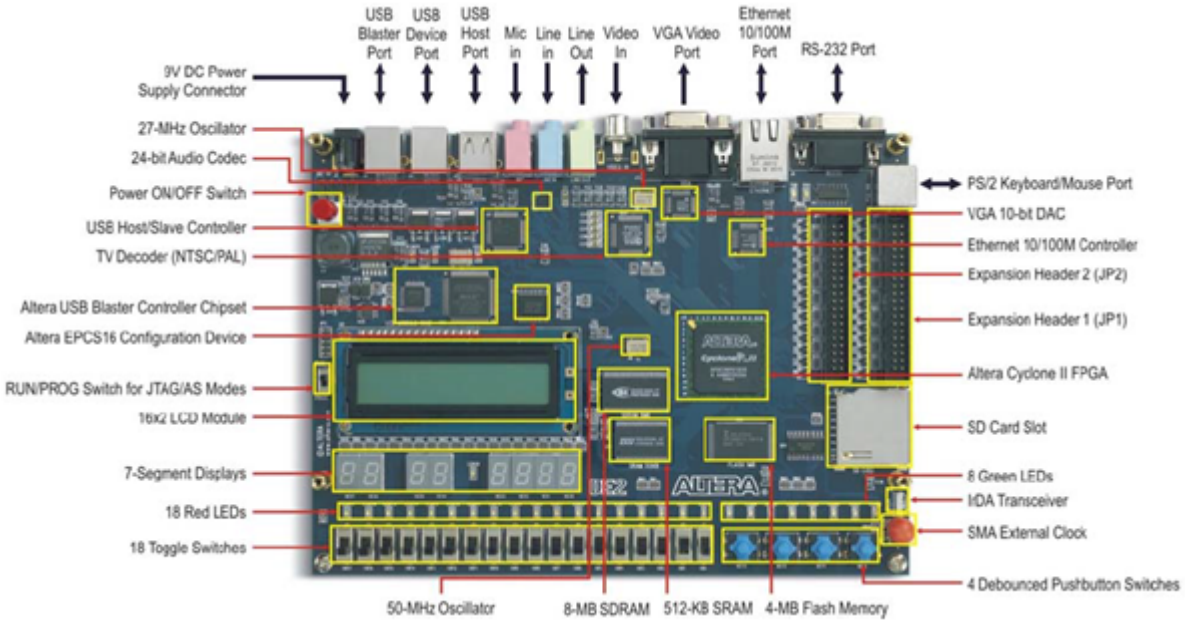
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Detailed specifications:

What does the system do?

The system is a two-player hang-man game.

We are going to implement a simultaneous symmetrical doble game of hang-man between two players (user and opponent). Each player is in front of a DE2 board. The two boards are connected with a serial cable using the 9 pin serial RS232 port. . The DE2 board is a development board from Altera, which is used to develop designs using the Altera Cyclone ® II 2C35 FPGA device.

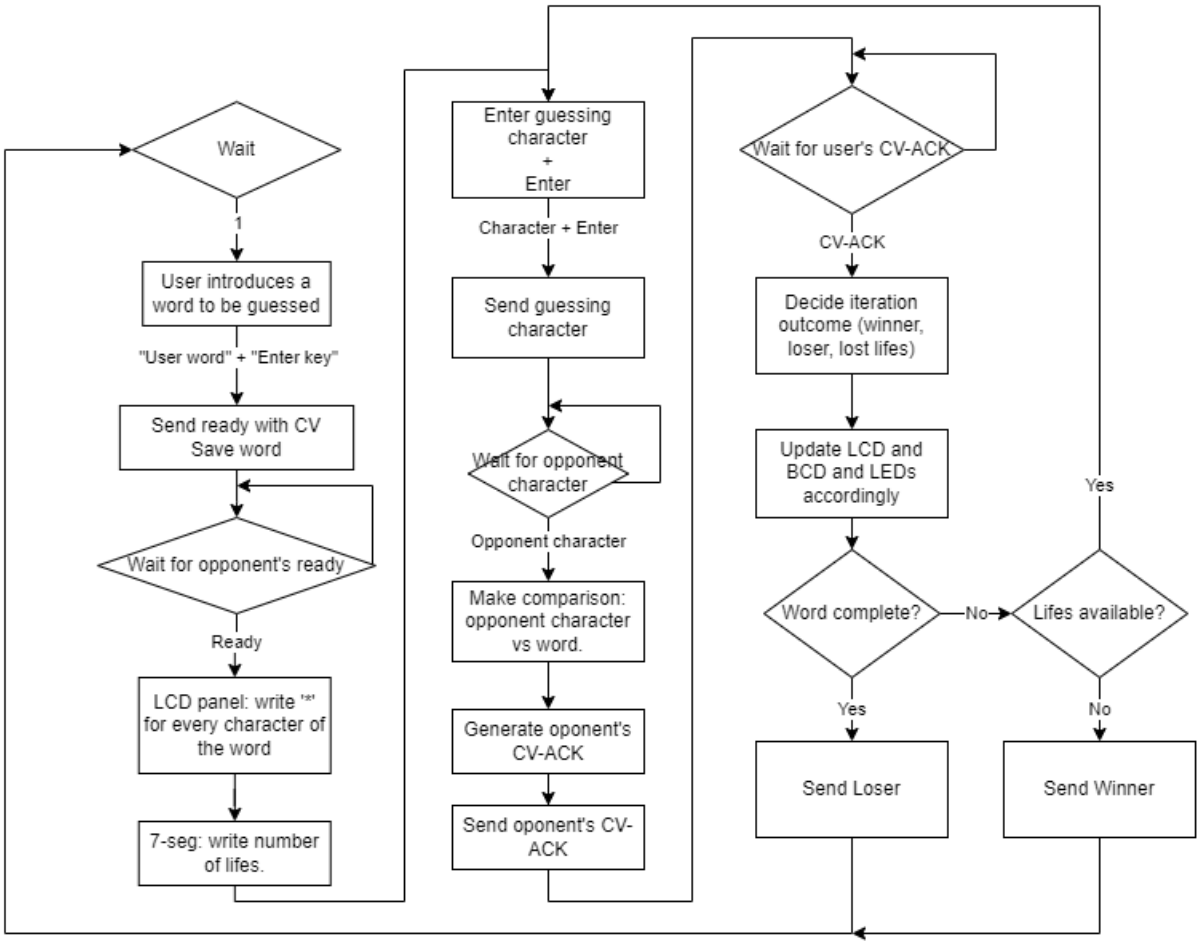


High level description of the algorithm

1. User: Presses 1 on the SP2 keyboard to start.
2. User: Introduces the word that needs to be guessed. *Note: No return (for the moment).*
3. User: Presses *enter* to confirm the word.
4. User: Sends the CV with ones in the used positions.
5. Opponent: send us his CV.
6. User: Introduces a guess letter. Enter to confirm.
7. Opponent: Send us the CV-ACK.
8. User:
    a. If the letter is in the word, the guessed letter is written in the LCD panel in the corresponding positions.
    b. If not, reduce the “lives” in the 7-segments and/or turn off a red LED.
9. Opponent: Same as the user.
10. Once one of the players has guessed all the characters present in the word, this is the winner.
    a. Winner:

i. All green LEDs on.
ii. 7-seg shows the word: WIN

b. Loser:
i. All red LEDs on.
ii. 7-seg shows the word: LOSE



Communications protocol:

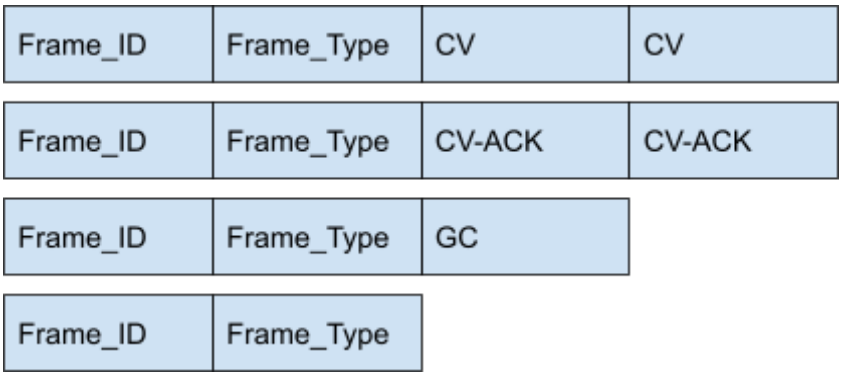
Physically, the boards will communicate through a serial cable, using 3 wires (RX, TX and GND) plugged to the 9 pin RS232 board connector.

The communication protocol is simple. The transmitter must build the frame, which is received and interpreted by the receiver. We will not implement any handshake. We assume that, as we are in a laboratory environment, there will be no communication errors. This means that once the sender has sent the frame, it assumes that it has been correctly received by the receiver. If there is a communication error, the system will not recognize the frame and the game must start over by pressing the RESET.

We design the following frames:

- Ready - Checking Vector (CV)
- Checking Vector ACK (CV-ACK)

- Guessing Character (GC)
- Loser
- Winner



FRAME_ID: The byte (AA)_hex
FRAME_TYPE:

Ready:	The byte (A1)_hex
CV-ACK:	The byte (B1)_hex
GC:	The byte (C1)_hex
Loser:	The byte (D1)_hex
Winner:	The byte (D2)_hex

DATA:

Ready:	2 bytes
CV-ACK:	2 bytes
GC:	1 byte
Loser:	0 bytes
Winner	0 bytes

The baud rate must be the same for the user and opponent boards. The input clock frequency is divided by 65536. A bit is going to be sent every 4 clock cycles.

The communication protocol is a ping-pong style. There are three moments while the game is in progress where there is communication between the two boards:

- To send the READY frame when the USER WORD has been introduced.
- To send the USER GUESSING CHARACTER.
- To send the result of the comparison between the USER GUESSING CHARACTER (CV-ACK) and the OPPONENT WORD.
- To inform the opponent that he/she has LOST.
- To inform the opponent that he/she has WON.

When one of these moments arrives, the board sends the adequate frame to the opponent board and then waits for the same frame from the opponent board. Communications are used to synchronize the games in the two boards: the moment when the game starts and the iterations. If one of the boards does not send the

frame, there is no time out and the game remains in a stand by mode. The only way to exit from this stand by mode is to RESET the system.

Option1:

At the beginning of the game, each player sends the word. Then when we receive a new character, the opponent knows where does it go in the word.

pros:

The user doesn't need to send the position of the letter → less communications.

cons:

We need to send an ACK anyway. And at the end we need to wait for the other player to finish and compare the lives available for each player.

Option2:

The word is never sent to the opponent. When the opponent sends a guess character, the user answers with a 2-byte vector with 1s in the position where it is. If the character is not present, the two-byte vector is all zero.

pros:

The ACK is the position vector of the guessed character.

Each player manages their own word.

cons:

There are more transmissions.

We finally decided for option two.

User interface:

Inputs:

- QWERTY PS2 keyboard to start the game ("1" key) and to introduce both the user word and the guessing character and their validation ("Enter" key).
- In the board:
    - Push button 1 (P1) to perform a global asynchronous reset.

Outputs:

- In the board:
    - RED LEDS (LR0…17): If all leds are ON, all the lifes are available. When a player runs out of lifes or loses the game all red leds turn OFF. For every life lost one led turns of starting for LR17 all the way to LR0.
    - GREEN LEDS (LG7…0): all leds are ON if the user wins the game.
    - The seven segments will show:
        - H0, H1, H2, H3: either " WIN" or "LOST" to indicate if the user has won or has lost the game.
        - H4, H3: will indicate the number of lifes remaining

- 16x2 LCD: Will show the word that the user is guessing. The characters that the user has guessed will appear at its corresponding position. The characters unguessed will show a hashtag.

Notes:
The LCD Pannel permits to write 16 columns and 2 rows.
The word is guessed character by character. No chance to guess the whole word.
16-bit vector with ones in the positions used for the word: CheckingVector (CV).